# Basic Game Theory

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What constitutes a "game" in mathematics?

- Two players (almost always, but not obligatory)
- Players affect the game state predictably, i.e. luck is not involved
- Players are fully aware of the state of the game
- Ending states in which one of the players wins are defined and are known to both players

Here are some properties that are not obligatory, but are often present:

- Players have access to the same move depending only on the state of the board (impartial game)
- Game lasts a finite number of moves (no draws)

The goal is to find out if both players play ideally, which player has a winning strategy.

**JBMO2023C3)** Alice and Bob play the following game on a $100 \times 100$ grid, taking turns, with Alice starting first. Initially, the grid is empty. At their turn, they choose an integer from 1 to $100^2$ that is not written yet in any of the cells and choose an empty cell, and place it in the chosen cell. When there is no empty cell left, Alice computes the sum of the numbers in each row, and her score is the maximum of these 100 numbers. Bob computes the sum of the numbers in each column, and his score is the maximum of these 100 numbers. Alice wins if her score is greater than Bob's score, Bob wins if his score is greater than Alice's score; otherwise no one wins.

Find if one of the players has a winning strategy, and if so, which player has a winning strategy.

**Divisor problem)** On the board are written numbers $1, 2, \ldots, n$. Players A and B colour the numbers in the following way: choose an uncoloured number $k$, colour it and all of the divisors of $k$. A player loses if he cannot make a move. If A plays first, which player has a winning strategy?

**ISL2009C1)** Consider 2009 cards, each having one gold side and one black side, lying parallel on a long table. Initially, all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing a block of 50 consecutive cards, the leftmost of which is showing gold, and turning them all over, so those which showed gold now show black and vice versa. The last player who can make a legal move wins.

(a) Does the game necessarily end?

(b) Does there exist a winning strategy for the starting player?

**SRB_OKR2022.4.5)** Let $m$ and $n$ be natural numbers. On the table are given two piles consisting of $m$ and $n$ gold coins. Players A and B play alternatively by choosing a gold pile and removing from it $d$ coins where $d$ is a divisor of the number of coins in the other gold pile. It is considered that 0 is divisible by all numbers. The winner is the player who takes the last coin. If A plays first, who has the winning strategy?

**SRB_DRZ2023.4.3)** On the board is written natural number $n$. Players A and B alternatively play the following move. If $m$ is written on the board, a player chooses a divisor $d$ of $m$ that is not 1 or $m$, erases $m$ and writes $m - d$ on the board. If A plays first, which player has a winning strategy?

**NIM game)** There are $n$ heaps of stones with $a_1, a_2, \ldots, a_n$ stones. Players A and B play a game where in each move they choose a non-empty heap and remove at least one stone from it. The player who can't make a move loses the game. Which player has a winning strategy?

**ISL2022C3)** In each square of a garden shaped like a $2022 \times 2022$ board, there is initially a tree of height 0. A gardener and a lumberjack alternate turns playing the following game, with the gardener taking the first turn:

-The gardener chooses a square in the garden. Each tree on that square and all the surrounding squares (of which there are at most eight) then becomes one unit taller. -The lumberjack then chooses four different squares on the board. Each tree of positive height on those squares then becomes one unit shorter.

We say that a tree is majestic if its height is at least $10^6$. Determine the largest $K$ such that the gardener can ensure there are eventually $K$ majestic trees on the board, no matter how the lumberjack plays.

**ISL2015C4)** Let $n$ be a positive integer. Two players $A$ and $B$ play a game in which they take turns choosing positive integers $k \leq n$. The rules of the game are:

(i) A player cannot choose a number that has been chosen by either player on any previous turn. (ii) A player cannot choose a number consecutive to any of those the player has already chosen on any previous turn. (iii) The game is a draw if all numbers have been chosen; otherwise the player who cannot choose a number anymore loses the game.

The player $A$ takes the first turn. Determine the outcome of the game, assuming that both players use optimal strategies.

**ISL2012C4)** Players $A$ and $B$ play a game with $N \geq 2012$ coins and 2012 boxes arranged around a circle. Initially $A$ distributes the coins among the boxes so that there is at least 1 coin in each box. Then the two of them make moves in the order $B, A, B, A, \ldots$ by the following rules: (a) On every move of his $B$ passes 1 coin from every box to an adjacent box. (b) On every move of hers $A$ chooses several coins that were not involved in $B$'s previous move and are in different boxes. She passes every coin to an adjacent box. Player $A$'s goal is to ensure at least 1 coin in each box after every move of hers, regardless of how $B$ plays and how many moves are made. Find the least $N$ that enables her to succeed.

**ISL2004C5)** $A$ and $B$ play a game, given an integer $N$, $A$ writes down 1 first, then every player sees the last number written and if it is $n$ then in his turn he writes $n+1$ or $2n$, but his number cannot be bigger than $N$. The player who writes $N$ wins. For which values of $N$ does $B$ win?